

Data Privacy: Final

June 11, 2026

ID : _____

Name : _____

1	/ 30
2	/ 30
3	/ 40
Total	/ 100

extra sheet

1. **Exact Score and Denoiser of a Two-Point Data Distribution** (30 points)

Let the data be the fair two-point distribution $X_0 \in \{+1, -1\}$, $\Pr[X_0 = +1] = \Pr[X_0 = -1] = \frac{1}{2}$. Fix one diffusion noise level and form the forward sample $Y = aX_0 + \sqrt{v}\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$ independent of X_0 , where $a = \sqrt{\bar{\alpha}} \in (0, 1)$ and $v = 1 - \bar{\alpha}$. Throughout, write the Gaussian density, the logistic sigmoid, and the hyperbolic tangent as

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad \sigma(t) = \frac{1}{1 + e^{-t}}, \quad \tanh(t) = \frac{e^t - e^{-t}}{e^t + e^{-t}}.$$

- (a) (5 points) Show that the marginal density of Y is the two-component mixture $p(y) = \frac{1}{2} f(y; a, v) + \frac{1}{2} f(y; -a, v)$.
- (b) (5 points) Let $r_+(y) = \Pr[X_0 = +1 \mid Y = y]$. Show $r_+(y) = \sigma\left(\frac{2a}{v}y\right)$, and hence the optimal (posterior-mean) denoiser is $\mathbb{E}[X_0 \mid Y = y] = \tanh\left(\frac{a}{v}y\right)$.
(Hint: $r_+/r_- = f(y; a, v)/f(y; -a, v) = e^{2ay/v}$; then $\mathbb{E}[X_0 \mid y] = r_+ - r_-$.)
- (c) (10 points) Show the exact score is $\partial_y \log p(y) = -\frac{y}{v} + \frac{a}{v} \tanh\left(\frac{a}{v}y\right)$, and verify consistency with (b) via the denoiser–score identity $\mathbb{E}[X_0 \mid Y = y] = \frac{1}{a} (y + v \partial_y \log p(y))$ (which you should also prove).
- (d) (10 points) Show separately that the denoiser $\mathbb{E}[X_0 \mid Y = y]$ converges to $\text{sign}(y)$ as $\bar{\alpha} \rightarrow 1$, and converges to 0 as $\bar{\alpha} \rightarrow 0$. Interpret each limit.

2. LiRA Closed Form and Decision Boundary (30 points)

For a target record z , the per-shadow-model loss is Gaussian under the two membership hypotheses:

$$\ell \mid (z \in D) \sim \mathcal{N}(\mu_{\text{in}}, \sigma_{\text{in}}^2), \quad \ell \mid (z \notin D) \sim \mathcal{N}(\mu_{\text{out}}, \sigma_{\text{out}}^2).$$

(a) (10 points) (Warm-up.) Derive the log-likelihood ratio

$$\log \Lambda(\ell) = \frac{(\ell - \mu_{\text{out}})^2}{2\sigma_{\text{out}}^2} - \frac{(\ell - \mu_{\text{in}})^2}{2\sigma_{\text{in}}^2} + \log \frac{\sigma_{\text{out}}}{\sigma_{\text{in}}}.$$

(Hint: subtract the two log-densities $\log \mathcal{N}(\ell; \mu, \sigma^2) = -\frac{(\ell - \mu)^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2)$.)

(b) (10 points) For $\sigma_{\text{in}} \neq \sigma_{\text{out}}$, show the region $\{\log \Lambda > \log \tau\}$ is a bounded interval, its complement, a half-line, all of $\mathbb{R}$, or empty, and say which from the sign of $\sigma_{\text{in}} - \sigma_{\text{out}}$. For equal variances $\sigma = \sigma_{\text{in}} = \sigma_{\text{out}}$ (and $\mu_{\text{in}} < \mu_{\text{out}}$), show it becomes a threshold $\mathcal{A}(z) = \mathbf{1}[\ell \leq c]$ and find c .

(Hint: $\log \Lambda$ is a quadratic $A\ell^2 + B\ell + C'$ with $A = \frac{1}{2}(\sigma_{\text{out}}^{-2} - \sigma_{\text{in}}^{-2})$; the sign of A sets the shape.)

(c) (10 points) With equal variances, derive the ROC: at false-positive rate α ,

$$\text{TPR}(\alpha) = \Phi\left(\Phi^{-1}(\alpha) + \frac{\mu_{\text{out}} - \mu_{\text{in}}}{\sigma}\right),$$

where $\Phi(t) = \int_{-\infty}^t \frac{1}{\sqrt{2\pi}} e^{-s^2/2} ds$ is the standard normal CDF.

(Hint: solve $\text{FPR}(c) = \Phi\left(\frac{c - \mu_{\text{out}}}{\sigma}\right) = \alpha$ for c , then substitute into $\text{TPR}(c) = \Phi\left(\frac{c - \mu_{\text{in}}}{\sigma}\right)$.)

3. Why NPO's Gradient Stays Bounded and GA's Does Not (40 points)

For LLM unlearning let $\pi_\theta(y | x) \in (0, 1]$ be smooth in θ , with $\pi_{\text{ref}}(y | x) \in (0, 1]$ a fixed reference. On the forget set $\mathcal{D}_F$ the two losses are

$$\mathcal{L}_{\text{GA}}(\theta) = \mathbb{E}_{\mathcal{D}_F}[\log \pi_\theta(y | x)], \quad \mathcal{L}_{\text{NPO}}(\theta) = \frac{2}{\beta} \mathbb{E}_{\mathcal{D}_F} \left[\log \left(1 + \left(\frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)} \right)^\beta \right) \right] \quad (\beta > 0),$$

where $\sigma(t) = 1/(1 + e^{-t})$ is the logistic sigmoid. Assume $\|\nabla_\theta \pi_\theta(y | x)\| \leq G$, and differentiate sums and expectations term by term freely.

(a) (10 points) Let $u(\theta) = \beta \log(\pi_\theta(y | x)/\pi_{\text{ref}}(y | x))$. Show that

$$\mathcal{L}_{\text{NPO}}(\theta) = \frac{2}{\beta} \mathbb{E}[\log(1 + e^u)], \quad \frac{\partial}{\partial u} \log(1 + e^u) = \sigma(u) \in (0, 1),$$

and that $\nabla_\theta u = \beta \nabla_\theta \log \pi_\theta$.

(Hint: $(\pi_\theta/\pi_{\text{ref}})^\beta = e^u$; $\frac{d}{du} \log(1 + e^u) = e^u/(1 + e^u) = \sigma(u)$, and $0 < \sigma < 1$ since $0 < e^u < 1 + e^u$; and π_{ref} is constant in θ .)

(b) (10 points) Using part (a) and the chain rule, show $\nabla_\theta \mathcal{L}_{\text{NPO}} = 2 \mathbb{E}[\sigma(u) \nabla_\theta \log \pi_\theta]$, and conclude the uniform bound $\|\nabla_\theta \mathcal{L}_{\text{NPO}}\| \leq 2 \mathbb{E}[\|\nabla_\theta \log \pi_\theta\|]$.

(Hint: differentiate $\frac{2}{\beta} \log(1 + e^u)$ with $\nabla_\theta u = \beta \nabla_\theta \log \pi_\theta$; then use the triangle inequality $\|\mathbb{E}[V]\| \leq \mathbb{E}[\|V\|]$ together with $\sigma(u) < 1$.)

(c) (10 points) Define the per-example NPO contribution $g_{\text{NPO}} := \sigma(u) \nabla_\theta \log \pi_\theta$. Show that

$$\|g_{\text{NPO}}\| = \sigma(u) \frac{\|\nabla_\theta \pi_\theta\|}{\pi_\theta}, \quad \sigma(u) = \frac{(\pi_\theta/\pi_{\text{ref}})^\beta}{1 + (\pi_\theta/\pi_{\text{ref}})^\beta},$$

and hence that as $\pi_\theta(y | x) \rightarrow 0$, $\|g_{\text{NPO}}\| \leq \frac{G}{\pi_{\text{ref}}^\beta} \pi_\theta^{\beta-1}$.

(Hint: $\nabla_\theta \log \pi_\theta = \nabla_\theta \pi_\theta / \pi_\theta$; bound $1 + (\pi_\theta/\pi_{\text{ref}})^\beta \geq 1$ to get $\sigma(u) \leq (\pi_\theta/\pi_{\text{ref}})^\beta$; combine with $\|\nabla_\theta \pi_\theta\| \leq G$.)

(d) (10 points) Using part (c), show that as $\pi_\theta \rightarrow 0$ the NPO per-example gradient $\|g_{\text{NPO}}\|$ stays bounded for $\beta \geq 1$, while the GA per-example gradient $\|\nabla_\theta \log \pi_\theta\|$ diverges. Why does this make NPO stable but GA collapse?

(Hint: read off the limits of $\pi_\theta^{\beta-1}$ from (c); the GA norm is $\|\nabla_\theta \pi_\theta\|/\pi_\theta$, whose numerator stays bounded while $\pi_\theta \rightarrow 0$.)

